

Aim

To configure and validate outbound firewall whitelisting for Qualys VMDR Scanner/Cloud Agent communication from a Kali Linux virtual machine by allowing approved Qualys IP addresses and CIDR ranges over TCP port 443.

Procedure**1. Check network connectivity**

```
ip addr
ip route
ping -c 4 8.8.8.8
nslookup google.com
```

2. Check and install UFW firewall

```
sudo ufw status
sudo apt update
sudo apt install ufw -y
sudo ufw status
```

3. Configure default firewall policies

```
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw enable
sudo ufw status verbose
```

4. Configure Qualys IP whitelisting

Allow HTTPS traffic to the approved Qualys IP addresses:

```
sudo ufw allow out to 168.138.113.116 port 443 proto tcp
sudo ufw allow out to 150.230.234.34 port 443 proto tcp
```

Allow the specified CIDR ranges:

```
sudo ufw allow out to 103.216.98.0/24 port 443 proto tcp
sudo ufw allow out to 103.75.173.0/24 port 443 proto tcp
```

Allow the specified Cloudflare IPs:

```
sudo ufw allow out to 162.159.152.21 port 443 proto tcp
sudo ufw allow out to 162.159.153.243 port 443 proto tcp
```

Verify the configured rules:

```
sudo ufw status numbered
```

5. Test Qualys DNS resolution

```
nslookup qualysguard.qg1.apps.qualys.in
nslookup distribution.qg1.apps.qualys.in
```

6. Test HTTPS connectivity

```
curl -I https://qualysguard.qg1.apps.qualys.in  
curl -I https://distribution.qg1.apps.qualys.in
```

7. Test TCP port 443 connectivity

```
nc -vz qualysguard.qg1.apps.qualys.in 443
```

If nc is not installed:

```
sudo apt install netcat-openbsd -y
```

8. Verify firewall rules

```
sudo ufw status numbered
```

9. Test the whitelisting by removing a rule

Delete one of the configured rules:

```
sudo ufw delete <rule-number>
```

Then test the Qualys connection:

```
curl -I https://qualysguard.qg1.apps.qualys.in
```

Restore the deleted rule after testing.

10. Check UFW firewall logs

```
sudo ufw logging medium  
sudo journalctl -k | grep UFW
```

Output

The following outputs were obtained during the experiment:

- Kali Linux network connectivity was verified using ip, ping, and DNS commands.
- UFW firewall was enabled with incoming connections denied and outgoing connections allowed.
- The specified Qualys IP addresses and CIDR ranges were added as outbound whitelist rules for TCP port 443 (HTTPS).